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Adaptive designs in public health: Vaccine and cluster randomized trials go Bayesian

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Clinical trials in public health—particularly those conducted in low- and middle-income countries—often involve communicable and non-communicable diseases with high disease burden and unmet needs. Trials conducted in these regions often are faced with resource limitations, so improving the efficiencies of these trials is critical. Adaptive trial designs have the potential to save trial time and resources and reduce the number of patients receiving ineffective interventions. In this paper, we provide a detailed account of the implementation of vaccine and cluster randomized trials within the framework of Bayesian adaptive trials, with emphasis on computational efficiency and flexibility with regard to stopping rules and allocation ratios. We offer an educated approach to selecting prior distributions and a data-driven empirical Bayes method for plug-in estimates for nuisance parameters.

KEYWORDS

Bayesian adaptive design, global health

1 | INTRODUCTION

There is a growing interest in adaptive trial designs for their data-driven approaches to clinical trial research.[1] In contrast to conventional (fixed sample) designs, where a single analysis is conducted at the end of the trial, adaptive trial designs use pre-specified interim analysis plans to modify aspects of an ongoing trial. Adaptive trial designs have the potential to save time and resources by reducing the overall number of patients required in a given clinical trial.[2]

While adaptive trial designs can be applied to all phases of clinical research, their application to public and global health has been limited.[3] Global health clinical trials have a specific focus on evaluating vaccines and other interventions that have the potential to improve health outcomes in low- and middle-income countries (LMICs), where the large burden of diseases and unmet public health needs exist. Vaccine trials and cluster randomized clinical trials are two common types of trials.[4,5] Vaccines against many important infectious diseases have contributed to infectious control and reduced global mortality rates.[6] Improvement of the development of novel vaccines is crucial to reduce the threat of future global health epidemics.

Cluster randomized clinical trials, where interventions are delivered at a group level, are often used in LMIC settings to evaluate public health intervention programs delivered at a mass level, not least because cluster randomization has the potential to reduce treatment contamination. Continuing to develop new adaptive trial design methods that can improve the quality and efficiency of these commonly used clinical trial designs in global health is of key interest.

The paper is organized in the following manner. In Section 2 we generalize event-driven Bayesian adaptive vaccine trials to any number of arms with any allocation ratio. We show that despite the statistical dependence between the different vaccine effects, vaccine efficacy can be inferred in the exact same manner as the standard two-arm trial. We

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provide guidance on prior distribution selection, and put it all together in an adaptive multi-arm, multi-stage vaccine trial design example.

Section 3 deals with Bayesian design and analysis of cluster randomized trials with count endpoints. We propose the Poisson-Gamma mixture (negative binomial) model as a means of modeling between-cluster variability, capitalize on the beta-negative binomial conjugation for efficient inference, tie the random effect parameters to the intracluster correlation coefficient and propose a simple empirical Bayes method for the derivation of a plug-in estimate for one of them. We then discuss some implications of putting the proposed methods to practice and wrap it all up in Section 4. Throughout, the methods developed are all efficient and MCMC-free, with vast numbers of simulations in mind.

2 | BAYESIAN MULTI-ARM, MULTI-STAGE VACCINE TRIALS

Double-blinded, randomized vaccine trials have been around since the famous, 1.3 million-large Polio vaccine trial,[7] but have come under increased public scrutiny recently, during the COVID-19 pandemic. The global state of emergency called for for innovative, more efficient designs,[8] and the Pfizer/BioNTech vaccine, trialed in a Bayesian group-sequential manner, received the U.S. Food and Drug Administration's emergency use approval (EUA) in record time.[9]

This section focuses on large, confirmatory trials designed to establish vaccine efficacy, where vaccine effect (VE) can be defined in a variety of ways, depending on the primary outcome of interest and the definition of susceptibility. Here we will discuss VEs that are not conditional on knowledge of exposure to infection.[10] The Bayesian framework for two-arm placebo-controlled trials with 1:1 randomization ratio has previously been discussed by Laurent & Legrand,[11] with the most prominent implementation to date being the phase III trial of the BNT162b2 COVID-19 vaccine.[12] For the rest of this section, we will extend the standard 2-arm model to any number of vaccine candidates with any randomization ratio, and will demonstrate the incorporation of this extended model in an adaptive trial scheme. Examples of ongoing adaptive, multi-arm vaccine trials include the PICOBOO trial[13] and the Solidarity Trial Vaccines,[14] both testing COVID-19 vaccine candidates.

2.1 | A Bayesian model

Suppose that we are trialing K new vaccines against a common control, such that the number of cases, $n_i(t)$, observed among the recipients of the ith vaccine t time units after the beginning of the trial follows a homogeneous Poisson process. For clarity: t does not represent the individual patient follow up time. Specifically, denoting by q_{ctrl} and q_i the allocation probabilities of the control and i^{th} active arm, respectively, and by μ_{ctrl} and μ_i the corresponding event rates within the trial follow up period, we may write

$$\begin{cases} n_{\text{ctrl}}(t) \sim \text{Poisson}(\lambda_{\text{ctrl}} t), \text{and} \\ n_i(t) \sim \text{Poisson}(\lambda_i t), i = 1 \ \dots \ , K, \end{cases}$$

where—denoting the relative rate reduction—or the *vaccine effect* (VE)—of the ith active vaccine group relative to control by v_i

$$\lambda_{\text{ctrl}} \propto q_{\text{ctrl}} \mu_{\text{ctrl}}$$

and

$$\lambda_i \propto q_i(1 - v_i)\mu_{\text{ctrl}}, i = 1, \ \dots \ , K$$

(assuming the same proportionality constant for all), where $q_{\text{ctrl}} + \sum_{j=1}^{K} q_j = 1$.

By time t, we have observed a total number of cases

$$n_{\text{total}}(t) = n_{\text{ctrl}}(t) + \sum_{j=1}^{K} n_j(t),$$

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where $n_{\text{ctrl}}(t)$ and $n_j(t)$ are the number of cases accumulated in the control group and the jth vaccine group by that same time, respectively, whose magnitude only becomes known upon unblinding the data. From the well known property of conditional independent Poisson processes given their sum[15] chap. 1,

$$n_1(t), \dots n_K(t), n_{\text{ctrl}}(t) | n_{\text{total}}(t) \sim \text{Multinom}(\theta_1, \dots, \theta_K, \theta_{\text{ctrl}})$$

for

$$\begin{aligned} \theta_i &= \frac{q_i(1-v_i)\mu_{\text{ctrl}} t}{q_{\text{ctrl}}\mu_{\text{ctrl}} t + \sum_{j=1}^{K} q_j\left(1-v_j\right)\mu_{\text{ctrl}} t} \\ &= \frac{q_i(1-v_i)}{q_{\text{ctrl}} + \sum_{j=1}^{K} q_j(1-v_j)} \end{aligned} \tag{1}$$

Inverting the linear system induced by (1) yields

$$v_i = 1 - \frac{\theta_i}{\theta_{\text{ctrl}}} \frac{q_{\text{ctrl}}}{q_i}. \tag{2}$$

Assigning $\boldsymbol{\theta} = (\theta_1, \dots, \theta_K, \theta_{\text{ctrl}})^{\mathsf{T}}$ a conjugate Dirichlet prior

$$\boldsymbol{\theta} \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_K, \alpha_{\text{ctrl}}), \tag{3}$$

we obtain the posterior distribution

$$\boldsymbol{\theta} | n_1(t), \dots n_K(t), n_{\text{ctrl}}(t) \sim \text{Dirichlet}(\alpha_1 + n_1(t), \dots, \alpha_K + n_K(t), \alpha_{\text{ctrl}} + n_{\text{ctrl}}(t)). \tag{4}$$

The posterior probability of efficacy for the ith vaccine at time t is then

$$p_i^{\text{effic}}(t) = \mathbb{P}(v_i > 0 | n_1(t), \dots n_K(t), n_{\text{ctrl}}(t)). \tag{5}$$

2.2 | Vaccine efficacy estimation and posterior inference

Often, (5) is approximated by drawing large Monte Carlo samples from the Dirichlet posterior (4). In Proposition 1, however, we offer a closed form expression that enhances computation. This proposition formally establishes what the reader may know intuitively: that any posterior inference on vaccine efficacy within the scheme proposed can be restricted to the vaccine arm of interest and the common control arm, irrespective of hyperparameters and allocation probabilities assigned to- or cases recorded in the other vaccine groups. This property makes the calculation of prior and posterior VE moments and quantiles exceptionally simple, as we proceed to show.

Proposition 1. *Define* $v_{K+1} := v_{\text{ctrl}}$ *with* $v_{K+1} \equiv 0$. *Then*

(*a*) *The joint posterior probability density function of the vaccine effects* $v_1, \dots, v_K$ *is given by*

$$p(v_1, \dots v_K | n_1(t), \dots n_K(t), n_{K+1}(t)) = \frac{\prod_{j=1}^{K+1} q_j^{\alpha_j + n_j(t)} \left(1-v_j\right)^{\alpha_j + n_j(t) - 1}}{\text{B}(\boldsymbol{\alpha} + \boldsymbol{n}(t)) \left[\sum_{j=1}^{K+1} q_j \left(1-v_j\right)\right]^{n_{\text{total}}(t) + \sum_{j=1}^{K+1} \alpha_j}}, \tag{6}$$

for $-\infty < v_j < 1, j = 1, \dots, K$, *where*

$$\text{B}(\boldsymbol{\alpha} + \boldsymbol{n}(t)) = \frac{\prod_{j=1}^{K+1} \Gamma\left(\alpha_j + n_j(t)\right)}{\Gamma\left(\sum_{j=1}^{K+1} \alpha_j + n_{\text{total}}(t)\right)}$$

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is the Beta function. In the special case of $K = 1$, (6) *reduces to*

$$p(v|n_{\text{vax}}(t), n_{\text{ctrl}}(t)) = \frac{q^{\alpha_{\text{vax}}+n_{\text{vax}}(t)}(1-q)^{\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t)}(1-v)^{\alpha_{\text{vax}}+n_{\text{vax}}(t)-1}}{\text{B}(\alpha_{\text{vax}}+n_{\text{vax}}(t), \alpha_{\text{ctrl}}+n_{\text{ctrl}}(t))(1-qv)^{\alpha_{\text{vax}}+\alpha_{\text{ctrl}}+n_{\text{vax}}(t)+n_{\text{ctrl}}(t)}}, \tag{7}$$

where q *is the allocation probability to the active vaccine arm.*

(b) *For all* $i = 1, \dots, K$,

$$p(v_i|n_1(t), \dots, n_K(t), n_{\text{ctrl}}(t)) = p(v_i|n_i(t), n_{\text{ctrl}}(t)).$$

This means that the individual posterior vaccine effects follow the distribution whose density is given in (7) *(as if it were a 2-arm trial).*

(c) *The* i^{th} *posterior mean vaccine effect* $(i = 1, \dots, K)$ *is given by*

$$\mathbb{E}[v_i|\boldsymbol{n}(t)] = 1 - \frac{q_{\text{ctrl}}}{q_i}\frac{\alpha_i + n_i(t)}{\alpha_{\text{ctrl}} + n_{\text{ctrl}}(t) - 1},$$

where the mean exists as long as $\alpha_{\text{ctrl}} + n_{\text{ctrl}}(t) > 1$ *(in particular, no prior mean exists for* $\alpha_{\text{ctrl}} \leq 1$*).*

(d) *The* ξ*-th posterior quantile of the* i^{th} *vaccine effect* $(i = 1, \dots, K)$ *is given by*

$$v_{i,\xi|\boldsymbol{n}(t)} = 1 - \frac{q_{\text{ctrl}}}{q_i}\frac{\theta_{\alpha_i+n_i(t),\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t),1-\xi}}{1-\theta_{\alpha_i+n_i,\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t),1-\xi}}, \tag{8}$$

where $\theta_{\alpha_i+n_i(t),\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t),1-\xi}$ *is the* $(1-\xi)$*-th quantile of the* $\text{Beta}(\alpha_i + n_i(t), \alpha_{\text{ctrl}} + n_{\text{ctrl}}(t))$ *distribution.*

(e) *The posterior probability of efficacy of the* i^{th} *vaccine* $(i = 1, \dots, K)$ *is given by*

$$p_i^{\text{effic}}(t) = F_{\alpha_i+n_i(t),\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t)}\left(\frac{q_i}{q_{\text{ctrl}} + q_i}\right), \tag{9}$$

where $F_{\alpha_i+n_i(t),\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t)}(\cdot)$ *is the Beta CDF with shape parameters* $\alpha_i + n_i(t)$ *and* $\alpha_{\text{ctrl}} + n_{\text{ctrl}}(t)$.

The proof of Proposition 1 is given in Appendix A.

Although it is standard in multi-arm trials to restrict attention to the relevant arms in paired comparisons, the dependence of the vaccine effects—as evident from (6)—meant that we needed the reassurance of part (b) in order to infer on each VE in isolation. The mathematical simplicity it equips us with, also allows us to explore prior distribution choices with ease, as we shall see next.

2.3 | Prior selection

While type I error rate control and power requirements of a simulation-guided adaptive design may be ensured via calibration of the decision rules (see Section 2.4), the choice of hyperparameters for the Dirichlet prior distribution (3) will still impact the vaccine effects estimation—in particular in an event-driven design like this, where case numbers are typically small.

For their famous COVID-19 vaccine trial,[12] suggested using $\alpha_{\text{vax}} = 0.700102$ and $\alpha_{\text{ctrl}} = 1$ (in a 1:1 allocation ratio trial), with the rationale that it should be centered at VE = 0.3—the World Health Organization's minimum requirement for the lower end of the resultant uncertainty interval—and was therefore considered "pessimistic". Senn[16] suggests constraining the hyperparameters to lie on contours that satisfy a prior mean requirement on the θ scale, lead to a unimodal prior and be as vague as possible.

We find the θ scale to be quite unintuitive, and suggest anchoring the choice of hyperparameter in the prior probability of efficacy. For example, if we wish to use "neutral" priors that assume an a priori 50/50 chance of vaccine efficacy, we may use (9) and find α_{vax} and α_{ctrl} that will satisfy

$$p_{\text{vax}}^{\text{effic}}(0) = F_{\alpha_{\text{vax}},\alpha_{\text{ctrl}}}\left(\frac{q_{\text{vax}}}{q_{\text{vax}} + q_{\text{ctrl}}}\right) = \frac{1}{2}, \tag{10}$$

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where $F_{\alpha_{\text{vax}},\alpha_{\text{ctrl}}}(\cdot)$ is the Beta CDF with shape parameters α_{vax} and α_{ctrl}. This is attainable by fixing α_{vax} at a small integer (that could be thought of as the number of infections observed in the vaccine group prior to the trial's start) and numerically finding α_{ctrl} as the root of (10).

Example 1. As a preparatory step for the adaptive design example of Section 2.4, we look to select hyper-parameters for the Dirichlet prior assigned to $\boldsymbol{\theta}$. The allocation ratio, as will be discussed in Section 2.4, is planned to be 1.11:1 in favor of the control arm, and the (normalized) allocation probabilities are therefore $(q_{\text{vax}}, q_{\text{ctrl}}) = (1/2.11, 1.11/2.11)$. For each $\alpha_{\text{vax}} \in \{1, 2, 5, 15\}$ we calculated its matching α_{ctrl} to warrant an a priori probability of efficacy of 0.5, and calculated some summaries for the resultant prior distribution, based on Proposition 1. The results are displayed in Table 1, as well as Figure 1. The choice of prior can thus be made using considerations such as its spread and inclusion of very high efficacy values in the uncertainty interval. In this example, we found $(\alpha_{\text{vax}}, \alpha_{\text{ctrl}}) = (1.00, 1.08)$ to be a good choice for a VE whose value we have little a priori idea of. Better informed VEs may be assigned more focused prior distributions.

2.4 | Adaptive vaccine trial design example

Consider a multi-arm, multi-stage trial designed to test the efficacy of three novel vaccine candidates against a placebo arm. To leave ourselves the option to stop vaccine arms early (potentially redistributing patients between the remaining arms) we plan three interim analyses and a final analysis. Interim analyses will take place when an average of 10 new diagnosed infection cases per a remaining arm have been recorded (that is: 40, 30 and 20 cases if four, three and two arms remain active, respectively), as well as one final analysis that will occur when either sufficiently many cases have piled up or the maximum follow up period has been reached.

The trial involves the comparison of multiple investigational vaccines against a common control. Although it is likely that in practice most principal investigators would go with a 1:1:1:1 allocation ratios, we would like to at least offer the

TABLE 1 Central and spread measures of the prior distribution candidates from Figure 1.

α_{vax}	α_{ctrl}	Mean	Median	95% CI
1.00	1.08	−13.21	0.00	[−31.84; 0.97]
2.00	2.18	−0.87	0.00	[−7.73; 0.89]
5.00	5.51	−0.23	0.00	[−2.55; 0.73]
15.00	16.59	−0.07	0.00	[−1.03; 0.51]

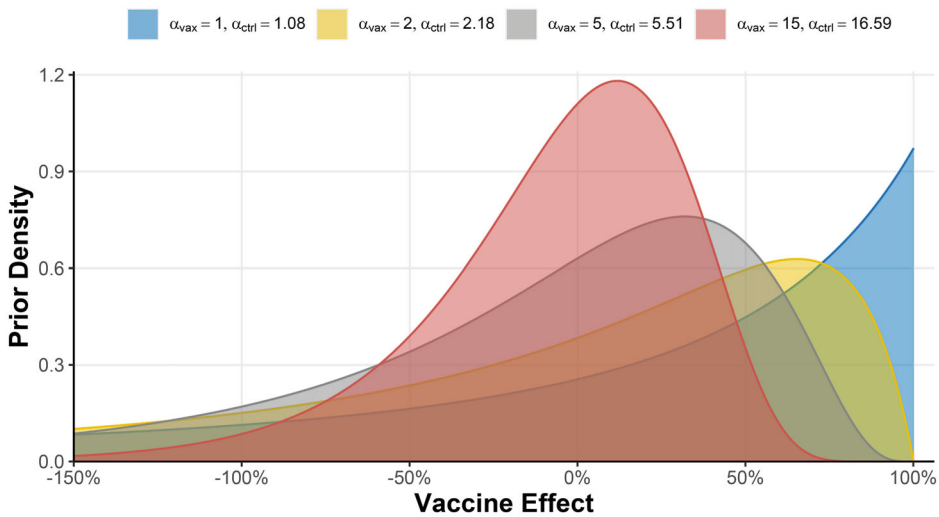

FIGURE 1 Prior distribution candidates for the vaccine trial for various $(\alpha_{\text{vax}}, \alpha_{\text{ctrl}})$ choices, as they appear in the vaccine effect scale.

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reader the possibility to explore other options. Even though the endpoint of interest in this case is not a dichotomous one, we recall that the Poisson distribution is a limiting case of the binomial distribution, and use it to motivate our choices. Suppose that we wished to compare the event rates in all investigational vaccine groups to the control group using the contrast $\boldsymbol{c} = \left[p_1 - p_{\text{ctrl}}, \ldots, p_K - p_{\text{ctrl}}\right]^{\top}$, using its estimator $\hat{\boldsymbol{c}} = \left[\hat{p}_1 - \hat{p}_{\text{ctrl}}, \ldots, \hat{p}_K - \hat{p}_{\text{ctrl}}\right]^{\top}$, whose covariance matrix is given by[17]

$$\text{Var}[\hat{\boldsymbol{c}}] = \text{diag}\left\{\frac{p_1(1-p_1)}{n_1}, \ldots, \frac{p_K(1-p_K)}{n_K}\right\} + \frac{p_{\text{ctrl}}(1-p_{\text{ctrl}})}{n_{\text{ctrl}}}\mathbf{1}\mathbf{1}^{\top}.$$

If we set $n_1 = \cdots = n_K = n$ and $n_{\text{ctrl}} = \rho n$, and choose ρ to be the optimizer of the *D-optimality* criterion, that is

$$\rho = \arg\min_{\rho^*} \det\{\text{Var}[\hat{\boldsymbol{c}}]\},$$

we may think of ρ (recalling the interpretation of the determinant as a volume function) as the control/treatment allocation ratio that minimizes the average variance across the different control-comparisons in some sense. Although this applies to a frequentist test, we know that Bayesian uncertainty intervals tend to be close to their frequentist counterparts, and may find this allocation ratio beneficial.

In this example, expecting 5% of the control arm patients to eventually get infected, and assuming a 37.5% relative rate reduction for the three vaccines, yielded a D-optimal 1.11:1:1:1 allocation ratio (in favor of the control arm). If arms get dropped along the way, the absolute allocation probabilities will change, but this 1.11:1 randomization probability ratio between control and active vaccine must persist for the analysis to remain valid.

Throughout the trial, we will be using the Dirichlet prior distribution with $\alpha_{\text{ctrl}} = 1.08$ and $\alpha_i = 1$, $i = 1, 2, 3$. At the jth interim readout ($j = 1, 2, 3$), the ith vaccine arm ($i = 1, 2, 3$) is

- Stopped for efficacy if $p_i^{\text{effic}}(j) > 0.99015$, as defined in (5).
- Stopped for futility if $p_i^{\text{effic}}(j) < 0.25j$. Note that the bar is raised in each analysis, to reflect the fact that there are fewer remaining observations that can cause a dramatic shift, or
- Advanced to the next analysis.

At the final analysis, efficacy testing takes place for the last time, using the same 0.99015 threshold on $p_i^{\text{effic}}(4)$. The reader may want to experiment with different schemas, such as using progressively more permissive efficacy thresholds with every interim analysis.

An example of a randomly simulated trial realization using the above adaptive design is given in Table 2. Here, the third vaccine arm was stopped early for futility after the second interim analysis, having crossed the 0.5 threshold on the posterior probability of efficacy downward. Of the remaining arms, the first one was stopped for efficacy one analysis later, while the second one continued until the trial's conclusion without any finding, having fallen just short of the 0.99015 threshold. These trajectories are documented in Figure 2a, while Figure 2b shows the eventual vaccine effect posterior density functions of the three arms. Note that each such control comparison is restricted to concurrent control data, which explains the increased uncertainty about the treatment effect of the third vaccine that was stopped early into the trial.

To evaluate the operating characteristics of the proposed design under specific conditions, a stochastic simulation was programmed and run for 100,000 iterations, as per the FDA's recommendation.[18] We conducted the simulation

TABLE 2 Bayesian analysis of the results of a random multi-arm, multi-stage vaccine trial simulation.

Arm	Stopping reason	Analysis 1 cases	Analysis 2 cases	Analysis 3 cases	Analysis 4 cases	Efficacy	ℙ(Efficacy|Data)	VE [95% CI]
Control		14	25	38	54			
1	End of trial	5	12	19	25	Yes	99.8%	48.0% [17.9%; 68.0%]
2	End of trial	7	19	29	37	No	90.1%	23.7% [−15.0%; 50.0%]
3	Early futility	14	24			No	41.5%	−6.2% [−84.8%; 39.1%]

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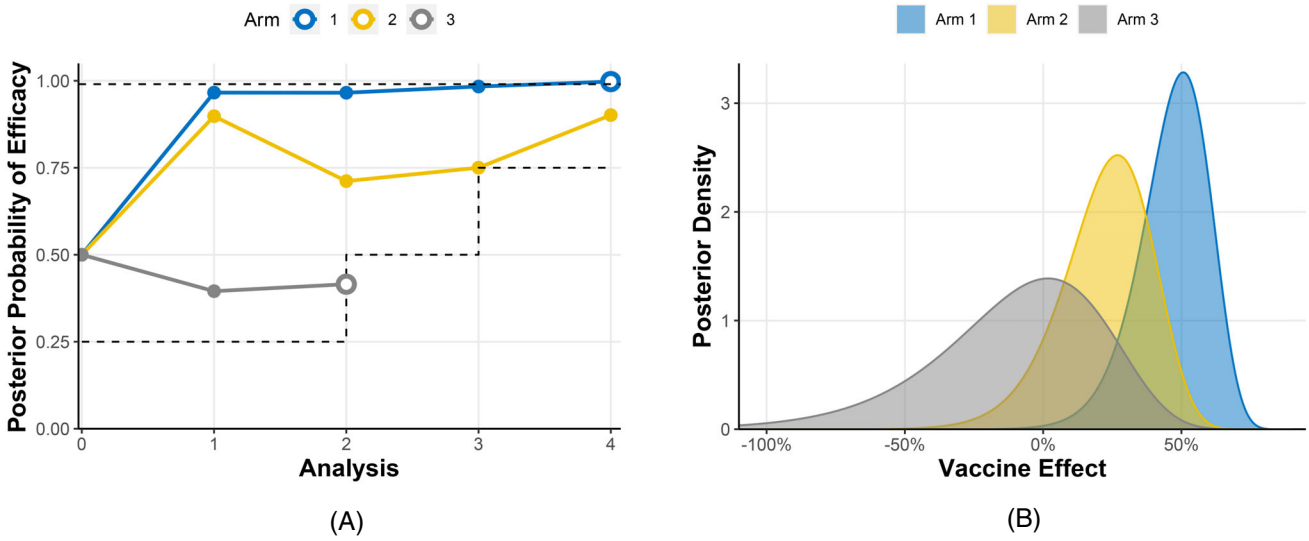

FIGURE 2 An example run. (a) Monitoring the posterior probability of efficacy over time for the multi-arm, multi-stage vaccine trial. (b) Posterior density plots of the vaccine effects of the three candidate vaccines relative to placebo.

TABLE 3 The operating characteristics of the proposed adaptive design of Section 2.4.

Arm	VE	Power	$\mathbb{P}$(Stop = 1)	$\mathbb{P}$(Stop $\leq$ 2)	$\mathbb{P}$(Stop $\leq$ 3)
All	0%	2.5%	23.3%	54.1%	78.9%
1	50%	80.9%	16.9%	42.7%	70.4%
2	37.5%	49.8%	10.2%	25.9%	50.7%
3	25%	22.9%	9.7%	23.9%	49.2%

under the assumption of three active vaccines with vaccine effects of 50%, 37.5% and 25%, respectively, all tested against a placebo arm, and in each iteration recorded the number of analyses each arm persisted for and whether it was declared efficacious according to the decision rules described earlier in this section. Additionally, an independent simulation was run under the "no effect" (null) scenario in which all VEs were set to zero, in order to evaluate the type I error rate.

The results are displayed in Table 3. In fact, the posterior efficacy threshold of 0.99015 was calibrated (through trial and error) to meet the common 2.5% type I error rate and 80% power (for the most efficacious arm) requirements. While the early stopping probabilities under both the 50% and the 0% VE hold promise for potentially considerable savings, although—as discussed in Section 4—these may not necessarily manifest themselves in the form of fewer doses administered.

Remark 1. Although we assumed for convenience homogeneous Poisson processes in Section 2.1, nothing in the results that followed will change if the intensity functions of said processes are nonhomogeneous, as long as the ratios between their respective intensity functions remain fixed over time.

3 | BAYESIAN ADAPTIVE CLUSTER TRIALS

Cluster randomized trial (CRT) designs are commonly used in health research when randomization at the individual level is not possible. Examples include group support psychotherapy,[19] maternal and child health,[20] and malaria prevention,[21] among others. Often, the intervention investigated does not allow blinding, although exceptions exist.[22] An important feature of CRTs is the compounding of the variability in the data, arising from the addition of a cluster-to-cluster variance component to the individual random error, leading to a potentially substantial inflation in the overall sample size

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requirements,[5] whose magnitude is governed by cluster size distribution, as well as the intracluster correlation coefficient (ICC),[23] capturing the level of similarity between observations from the same cluster. This inefficiency of CRTs relative to individually randomized trials, means that we only ever use them when we have no other choice, as in the example of malaria prevention via the use of attractive toxic sugar baits (ATSBs), where placing a bait outside the village impacts its entire population.

In this section we propose a Bayesian model for the analysis of a cluster randomized clinical trial with a count primary endpoint. We combine rigor with pragmatism to enable the use of conjugate prior distributions to enhance computational efficiency, and demonstrate how this model can be used within an adaptive trial design. Previous examples of simulation-guided Bayesian adaptive cluster trials involved MCMC sampling within simulations.[24] Note that unlike the event-driven vaccine trial of Section 2, this trial is patient-driven (ie, analyses are triggered once a certain number of clusters have seen out their treatment), and follow-up times are assumed to be fixed.

3.1 | Count outcome model

Assume (for now) that all clusters randomized to treatment arm $i = 1, \dots I$ are of the same size n_i, and denote by r_{ij} the event rate in the jth cluster ($j = 1, \dots K_i$) of the ith treatment arm. The Poisson distribution is then a popular choice for modeling the number of cases observed within the trial's follow up period, that is,

$$Y_{ij} | r_{ij} \sim \text{Poisson}\left(n_i r_{ij}\right).$$

To capture the cluster-to-cluster incidence rate variability, we may use a random effect model, namely

$$r_{ij} | \mu_i, \phi_i \sim \text{Gamma}(\phi_i, \mu_i/\phi_i), \tag{11}$$

where μ_i in (11)—now the mean incidence rate of the ith treatment group—becomes our main parameter of inferential interest, and ϕ_i is a *shape* parameter that we assume to be known at this point. Some degree of caution is required, as one must confirm that the combination of μ_i and ϕ_i makes it practically impossible for r_i to exceed 1 within a simulation setting. While we apply truncation at 1 to ensure the robustness of the simulation, we do not want this to be of any practical implications. Then, with some effort, it can be shown[25] chap. 1 that

$$\mathbb{P}\left(Y_{ij} = y_{ij} | p_i, \phi_i \right) = \frac{\Gamma\left(y_{ij} + \phi_i\right)}{\Gamma(\phi_i) y_{ij}!} p_i^{y_{ij}} (1 - p_i)^{\phi_i}, \tag{12}$$

for

$$p_i = \frac{\mu_i n_i}{\phi_i + \mu_i n_i}. \tag{13}$$

The implication of (12) is that marginally (when averaging out cluster heterogeneity), the number of cases per cluster in each treatment arm will follow a negative binomial distribution. All inferences on μ_i may be drawn by re-transforming (13) via the relationship

$$\mu_i = \frac{\phi_i}{n_i} \frac{p_i}{1 - p_i}. \tag{14}$$

This proves to be of crucial importance, as a conjugate Beta(a_i, b_i) prior distribution may now be assigned to p_i, yielding

$$\begin{aligned} p\left(p_i | \boldsymbol{y}_i, \phi_i\right) &\propto \left\{ \prod_{j=1}^{K_i} \mathbb{P}\left(Y_{ij} = y_{ij} | p_i, \phi_i \right) \right\} \pi(p_i) \\ &= p_i^{a_i - 1} (1 - p_i)^{b_i - 1} \times p_i^{\sum_{j=1}^{K_i} y_{ij}} (1 - p_i)^{K_i \phi_i} \\ &= p_i^{a_i + \sum_{j=1}^{K_i} y_{ij} - 1} (1 - p_i)^{b_i + K_i \phi_i - 1}, \end{aligned} \tag{15}$$

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Where K_i is the number of clusters randomized to the ith treatment arm. From (15) it is clear that

$$p_i|\boldsymbol{y}_i, \phi_i \sim \text{Beta}\left(a_i + \sum_{j=1}^{K_i} y_{ij}, b_i + K_i\phi_i\right). \tag{16}$$

We can then take advantage of (16) and (14) to draw large Monte Carlo samples from the posterior distribution of μ_i, $i = 1, \ldots, K$, in order to compare the different treatment arms, without the need to utilize Markov Chain Monte Carlo methods[26] within each trial simulation, which could require a significant amount of computing resources for the number of simulations required to accurately evaluate the design's operating characteristics. When it is time to analyze the data collected throughout the trial, we may very well need to resort to running MCMC for for example, covariate adjustment, but at the design stage, where numerously many simulations must be run for a variety of scenarios, speed is key.

3.2 | Prior elicitation

In order to simulate a Bayesian adaptive CRT based on the above principles, value for the different shape parameter ϕ_i must be supplied to the Gamma random effect (11). Ideally, these values should reflect an interpretable statistical aspect of the data generating mechanism. Proposition 2 ties ϕ_i to the ith treatment arm's ICC ρ_i, in a way that will facilitate plugging in plausible ϕ values. Furthermore, it provides guidance with regard to choosing Beta prior distributions for the transformed parameters p_i in a meaningful way, through the resultant prior distribution assigned to the main parameters of interest, μ_i.

Proposition 2. *The following holds for the Bayesian cluster trial analysis model of Section 3.1*

(*a*) *The ICC parameter ρ_i for the ith treatment arm and the corresponding shape parameter ϕ_i of the Gamma event rate random effect are related via*

$$\rho_i = \frac{\mu_i}{\phi_i(1-\mu_i)} \tag{17}$$

or conversely,

$$\phi_i = \frac{\mu_i}{\rho_i(1-\mu_i)}. \tag{18}$$

(*b*) *If we assign a prior distribution $p_i \sim \text{Beta}(a_i, b_i)$, then the posterior mean of the corresponding mean event rate parameter μ_i will be given by*

$$\mathbb{E}\left[\mu_i|\boldsymbol{y}_i, \phi_i\right] = \frac{\phi_i\left(a_i + \sum_{j=1}^{K_i} y_{ij}\right)}{n_i(b_i + K_i\phi_i - 1)}, \tag{19}$$

where the mean exists as long as $b_i + K_i\phi_i > 1$ (in particular, no prior mean exists for $b_i \leq 1$).

The proof of Proposition 2 is given in Appendix B.

Example 2. Suppose that we plan to trial three treatments against each other in a cluster randomized design with cluster size $n = 50$, and look to evaluate a proposed design under the assumption that the three arms have mean event rates of 10%, 7.5% and 5%. From past studies, it appears that the typical ICC for the primary outcome is $\rho = 0.15$. Using (18), we learn that the matching shape parameters for the three treatment arms are $\phi = 0.74$, 0.54 and 0.35, respectively.

The second half of Proposition 2 hints at one possible choice for a Beta(a, b) prior for the p parameters in (13). From (19), we gather that for the prior distribution of the mean event rate μ to be centered at mean $\mathbb{E}[\mu]$, we have to choose

$$b = 1 + \frac{\phi a}{n\mathbb{E}[\mu]}, \tag{20}$$

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We may thus assume a value for $\mathbb{E}[\mu]$, calculate its corresponding ICC ρ from (18) as in Example 2, and proceed to find the matching b hyperparameter values for a list of a values.

Example 3. To build on Example 2, we now wish to postulate Beta prior distributions for p_1, p_2 and p_3. We intend to assume no a priori differences between the three treatments by assigning identical priors, that will be centered—on the mean event rate scale—at 10%. Using the $\phi = 0.74$ value that was calculated in Example 2 in (20), and experimenting with $a = 2$, 4, 6 and 8, we obtain $b = 1.30$, 1.59, 2.19 and 3.37, respectively. The resultant priors on the μ scale are presented in Figure 3, with their 95% credible intervals appearing in Table 4.

3.3 | Shape parameter tuning via empirical Bayes

In addition to generating random data sets, values for the shape parameters ϕ_i must be provided in order to infer on the mean event rates μ_i, through (14). Using the true ϕ_i values in the analysis would lead to operating characteristics that flatter to deceive. We therefore need to devise a strategy for deriving plug-in ϕ_i estimates, if we are to avoid the assignment of prior distributions $\pi(\phi_i)$ and subsequently running many thousands of slow Markov Chain Monte Carlo (MCMC) iterations within each of the 100,000 simulated trials.

One popular approach for obtaining plug-in estimates is *Empirical Bayes*,[27] where the likelihood function is averaged over all parameters of inferential interest with respect to their prior distribution. To this end, we denote by $\mathcal{L}_i(\phi_i)$ the marginal likelihood based on the data emerging from the ith treatment arm. Integrating with respect to $\pi(p_i)$ yields

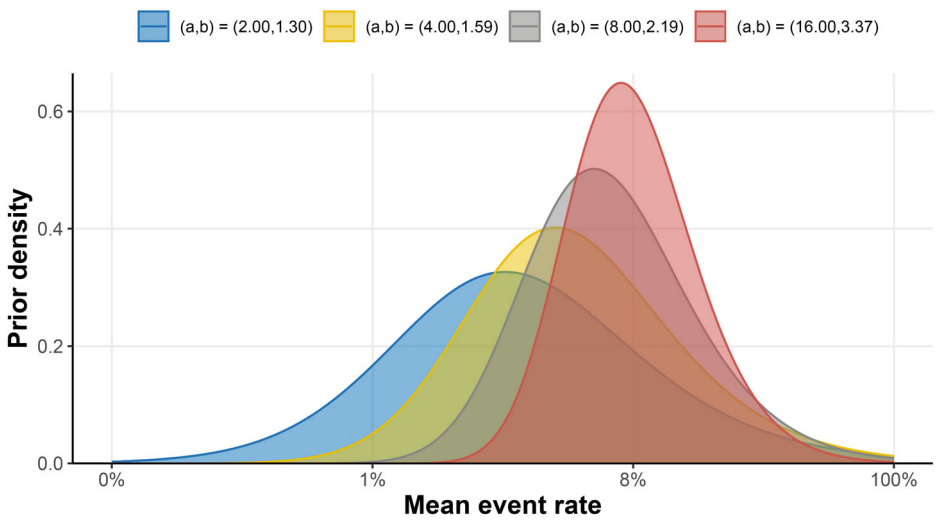

FIGURE 3 Prior distribution candidates for a Bayesian cluster trial for various (a, b) choices, as they appear in the mean control event rate scale.

TABLE 4 Candidate Beta prior distributions for Bayesian cluster trial analysis.

$\mathbb{E}[\mu]$	ρ	ϕ	(a, b)	95% CI
0.10	0.15	0.74	(2.00, 1.30)	[0.22%; 46.29%]
			(4.00, 1.59)	[0.70%; 48.32%]
			(8.00, 2.19)	[1.49%; 41.25%]
			(16.00, 3.37)	[2.56%; 31.67%]
	0.20	0.56	(2.00, 1.22)	[0.18%; 42.06%]
			(4.00, 1.44)	[0.56%; 48.23%]
			(8.00, 1.89)	[1.24%; 44.36%]
			(16.00, 2.78)	[2.21%; 35.14%]

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$$\begin{aligned}\mathcal{L}_i(\phi_i) &= \int_0^1 \prod_{j=1}^{K_i} \mathbb{P}\left(Y_{ij} = y_{ij} \middle| p_i, \phi_i\right) \pi(p_i) \mathrm{d}p_i \\ &= \left[\prod_{j=1}^{K_i} \frac{\Gamma\left(y_{ij} + \phi_i\right)}{\Gamma(\phi_i) y_{ij}!}\right] \int_0^1 p_i^{a_i + \sum_{j=1}^{K_i} y_{ij} - 1} (1 - p_i)^{b_i + K_i \phi_i - 1} \mathrm{d}p_i \\ &\propto \frac{\mathrm{B}\left(a_i + \sum_{j=1}^{K_i} y_{ij}, b_i + K_i \phi_i\right) \prod_{j=1}^{K_i} \Gamma\left(y_{ij} + \phi_i\right)}{[\Gamma(\phi_i)]^{K_i}},\end{aligned}$$

and after applying the logarithmic transformation and collapsing all the terms that do not involve ϕ_i, we obtain the log-marginal likelihood

$$\ell_i(\phi_i) = \log \mathrm{B}\left(a_i + \sum_{j=1}^{K_i} y_{ij}, b_i + K_i \phi_i\right) + \sum_{j=1}^{K_i} \log \Gamma\left(y_{ij} + \phi_i\right) - K_i \log \Gamma(\phi_i) + \text{const.} \tag{21}$$

Simple numerical optimization of the individual log-marginal likelihood (as univariate functions of the ϕ_i's) at the respective arm-level datasets will then give us the empirical Bayes estimates

$$\hat{\phi}_i = \arg\max_{\phi_i} \ell_i(\phi_i), i = 1 \ldots I,$$

that will be treated as fixed constants thereafter. Proposition 3 assures us of the existence of a local maximum. Although multiple local maxima have not been observed at any point in the simulations run for this paper, those should be easily handled using a multi-start approach. The proof of Proposition 3 is given in Appendix C.

Proposition 3. *For $a_i \in \mathbb{N}^+$, the log-marginal likelihood (21) has at least one local maximum.*

In a full, 100,000-long trial simulation for the evaluation of a design's operating characteristics, we will incorporate this empirical Bayes procedure at each simulated interim readout of the data.

Figure 4 shows the log-marginal likelihood curves for the data given in Table 5, with both the true shape parameter values used for generating the data and their empirical Bayes estimates.

3.4 | Adaptive cluster randomized trial design example

We now proceed to utilize everything discussed up to this point in this section in a Bayesian adaptive cluster randomized trial design. Suppose now that we wish to trial the three treatments of Example 2, with a planned maximum

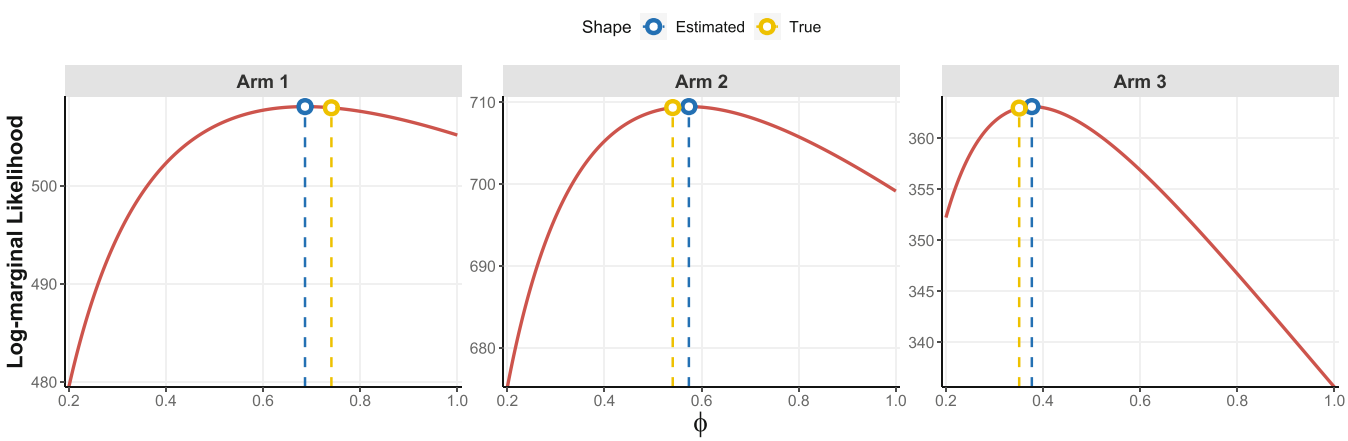

FIGURE 4 Final empirical Bayes estimates of the shape parameters from the randomly simulated cluster trial from Section 3.4, whose data are given in Table 5, obtained by maximizing the individual arm log-marginal likelihoods.

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TABLE 5 Bayesian analysis of a single simulation of the cluster trial design of Section 3.1.

Arm	Analysis 1 Clusters, Cases/Patients	Analysis 2 Clusters, Cases/Patients	Analysis 3 Clusters, Cases/Patients	Outcome	Mean event rate [95% CI]
1	62, 261/3075	123, 559/6203		Early inferiority	8.9% [7.2%; 11.3%]
2	62, 202/3057	123, 469/6022	213, 844/10558		7.9% [6.6%; 9.6%]
3	62, 206/3059	123, 341/6186	213, 582/10652	Superiority	5.4% [4.3%; 6.9%]

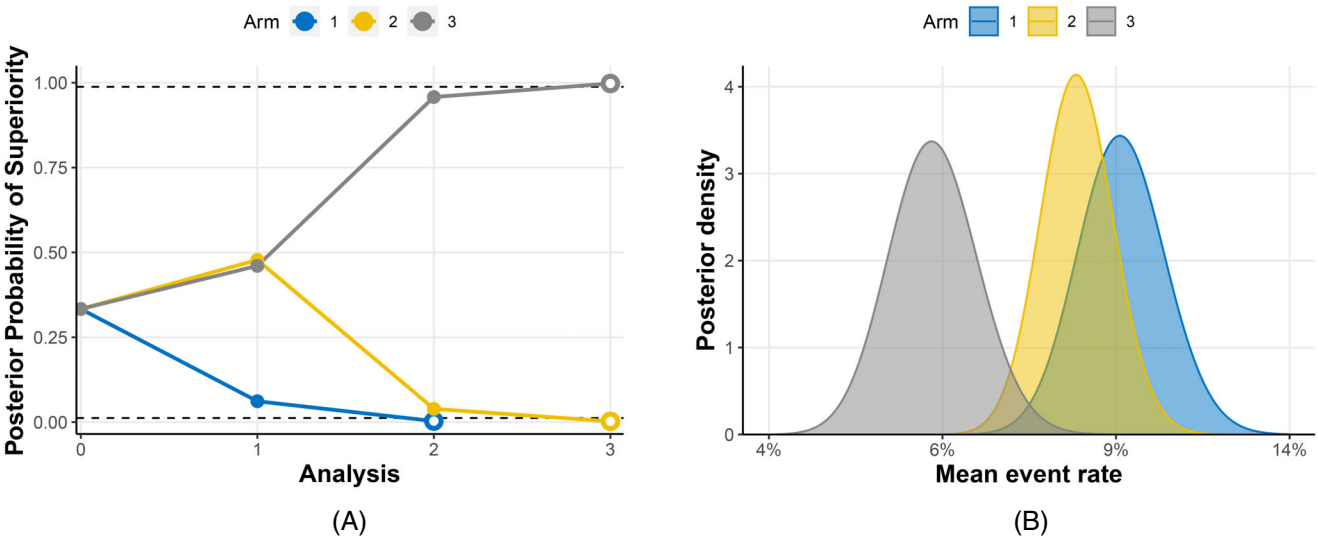

FIGURE 5 Bayesian analysis of the single trial simulation of Table 5. (a) The posterior probability of superiority by analysis. (b) Posterior density plots for the three mean event rate parameters at the end of the trial.

sample of 550 cluster with an average size of 50 patients each (in the simulation we sample the cluster sizes from a Poisson(50) distribution. While this number of clusters may seem unimaginable, larger CRTs have been conducted.[28] We plan two interim analyses and a final analysis. Unlike the vaccine trial of Section 2, we do not wish to perform multiple control-comparisons, but rather determine whether any of the three treatments is superior to the two others, and identify it. For this purpose, we define the ith arm *posterior probability of superiority*

$$p_i^{\text{sup}} = \mathbb{P}(\mu_i = \min\{\mu_1, \mu_2, \mu_3\}|\boldsymbol{y}), i = 1, 2, 3, \tag{22}$$

that will serve as our test statistic for decision making. Throughout the trial, we will be using the Beta(2.00, 1.30) prior distribution that is centered at the 10% mean event rate and in Example 3 seemed to strike a good balance between prior information and ignorance.

At any interim analysis, the ith treatment arm ($i = 1, 2, 3$) is stopped for inferiority if $p_i^{\text{sup}} < 0.012$, as defined in (22). The remaining arms continue until for any of them $p^{\text{sup}} \geq 0.988$, in which case it is announced as the most efficacious treatment and the trial stops. At the final analysis, efficacy testing takes place for the last time, using the same 0.988 threshold on p^{sup}. An example run is showed in Table 5 and Figure 5.

The authors recognize that interim analyses that take place upon treatment completion at the cluster level are rare and require special circumstances (namely: short follow up times). One such example was the EPIC-PEP trial,[29] testing the prophylactic use of Paxlovid® in household members of index COVID-19 cases. Note that this is not an example of a trial we simulate, due to the small clusters—formed by households—as well as the conditionally dependent observations within each cluster (as a result of the infectious nature of the disease).

To calculate some operating characteristics (under certain assumptions) for this proposed design, we again ran a full, 100,000-long Monte Carlo simulation under both null and non-null scenarios, assuming again an ICC of 0.15 and using empirical Bayes for plug-in values of the shape parameters ϕ_i within each simulated trial. Here, two "null" scenarios

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TABLE 6 Operating characteristics for the adaptive cluster trial design of Example 3.1.

Mean event rates	Mean no. of clusters	% of patients in Arm 1	% of patients in Arm 2	% of patients in Arm 3	TPR	FPR
(10.0%, 7.5%, 5.0%)	412.1	20.9%	38.8%	40.3%	80.6%	0.0%
(10.0%, 10.0%, 10.0%)	545.7	33.3%	33.3%	33.3%	—	2.0%
(10.0%, 5.0%, 5.0%)	532.0	13.0%	43.5%	43.5%	—	9.7%

were considered: one in which all arms had a 10% event rate, and another in which two treatments offered an identical benefit compared to the third one (5% vs. 10%). The "non-null" scenario included a clear hierarchy, in which the third treatment had the lowest event rate of 5%, compared to 7.5% event rate of the second one and 5% of the first treatment. Table 6 summarizes our findings. The reader should note that more complex hypotheses are being tested here: the "null hypothesis" consists not only of equipoise between all three arms (second row), but rather, that no treatment is the best of the three (third row). "True positive rate" (TPR) here is therefore the probability of ruling out equipoise and identifying the correct treatment among the three (should one exist), while "False positive rate" (FPR) is the probability of incorrectly declaring a winner.

Although the proposed design is purely hypothetical (more on this in Section 4), Table 6 provides some hints of the potential benefits of an adaptive, simulation-guided design. Basic sample size calculations for a two-arm CRT with variable cluster size[30] suggest that if we were to conduct two individual comparisons between Arm 1 and Arm 2 and Arm 2 and Arm 3 at the 5% level (one-sided), with no multiplicity adjustment, and powered each to the 80% level without accounting for the variable cluster size, the 1–2 comparison alone would require a total of 528 clusters. Moreover, under the "alternative" scenario (top row), the probability of clusters receiving the inferior treatment plummets to 20.9%—although we must ensure that sufficiently many clusters have received each treatment before any adaptation is made, to reduce the risk of covariate imbalance. Although 9.7% may seem high (third row), it must be taken into account that this FPR was calculated under the most extreme scenario of two equally effective treatments, where the risk of a false positive is likely minimal.

4 | PRACTICAL CONSIDERATIONS AND CONCLUDING REMARKS

With potential gains in sample size requirements, number of futile doses administered and time to regulatory approval, the lure of adaptive trial design may appear hard to resist. Here we would like to discuss some of the challenges that may emerge with the choice of a more logistically-demanding design.

In both Sections 2.4 and 3.4 we proposed designs with early stopping rules at interim analysis. When including such rules, the principal investigator must consider (a) the interplay between treatment duration/time to outcome and enrollment rate, and (b) the benefit of early stopping. In a vaccine trial, for example, it is quite possible the by the time that sufficiently many symptomatic infections have been accrued to trigger an interim analysis, all trial participants have been enrolled, randomized and had a vaccine or placebo dose administered. Early stopping here would therefore not save any resources, but may rather serve to apply earlier for EUA by the regulator, or unblind patients from a futile arm and prevent them from triggering future interim analyses—and possibly offer them an efficacious vaccine dose.

Group sequential designs such as the adaptive CRT of Section 3.4 may not be feasible if the primary outcome requires years of follow up,[31] in which case the investigator may resort to common control comparisons as in the proposed vaccine trials, or even consider the use of a quicker-to-observe, surrogate outcome.[32] Another ramification of prolonged trials is the inevitability of including a temporal term in the outcome model, to anticipate time-varying event rates, that result in pooling together clusters from non-overlapping time periods

Arguably the greatest savings in time and resources are offered through the incorporation of multiple prospective therapies in a *platform trial*,[33,34] where treatments may be dropped and added over time, and a single, pre-approved *master protocol* replaces individual-therapy trial protocols (reduced to Intervention-specific Appendices (ISA)). To conduct an event-driven trial such as the vaccine trial of Section 2.4, the investigator must be mindful of any antibody levels ramp up period required, and start administering any newly added vaccine ahead of time. Likewise, any waning immunity in ongoing vaccines must be addressed by administering a booster dose.

Response adaptive randomization (RAR),[1] whereby allocation probabilities are allowed to change dynamically to favor treatments that appear more promising throughout the trial. They may increase the probability of identifying the overall

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best treatment in a multi-arm, multi-stage trial at the expense of pairwise comparisons, and present the clinical research associates (CRAs) with nontrivial logistical challenges.[35] In this paper we made no use of RAR in Section 2.4, simply because the requirement that the vaccine:placebo allocation ratio be kept fixed throughout rules it out. While no clear-cut prohibition on the use of RAR in adaptive cluster trials exists, the reader is warned again that these trials are more prone to covariate imbalance between arms, and all adaptations must be carefully considered.

The approaches discussed in this paper can clearly be built upon and extended. Time-varying vaccine effects[10] have been studied for many years and are likely to provide a more accurate approximation of reality than the fairly naive model of Section 2. Stepped wedge cluster randomized trials[30] may be a natural extension of Section 3 through the inclusion of a term capturing the treatment period of each cluster. Bayesian Cluster randomized trial designs for count and time to event endpoints will require their own implementation details, as a closed-form posterior distribution for numeric outcomes has been derived by Shen et al.[24] We trust that with the growing interest in adaptive trial designs, the intensity of the research in the field of public and global health clinical trials will increase in the forthcoming years.

CONFLICT OF INTEREST STATEMENT

The authors declare having no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study. The analyses appearing in this paper were carried out using version 4.3.0 of the R statistical software.[36] R code, along with an online tutorial reviewing all methods and examples, can be accessed at https://github.com/oharari/Bayesian-Adaptive-Public-Health-Designs.

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APPENDIX A. PROOF OF PROPOSITION 1

(a) Recall that

$$\begin{aligned}\theta_i &= \frac{q_i(1-v_i)}{q_{\text{ctrl}} + \sum_{j=1}^{K} q_j\left(1-v_j\right)} \\ &= \frac{q_i(1-v_i)}{\sum_{j=1}^{K+1} q_j\left(1-v_j\right)}, i = 1, \ldots, K.\end{aligned} \tag{A1}$$

Hereafter, we shall denote for convenience $q_{K+1} = q_{\text{ctrl}}$ and $v_{K+1} = 0$. The Jacobian matrix of the transformation (A1) can be shown to be

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$$\boldsymbol{\mathcal{J}} = \frac{\partial \boldsymbol{\theta}}{\partial \boldsymbol{v}} = \frac{-1}{\sum_{j=1}^{K+1} q_j\left(1-v_j\right)} \boldsymbol{\mathcal{Q}}\left(\mathbb{I} + \boldsymbol{u}\mathbf{1}^{\mathsf{T}}\right)$$

for

$$\boldsymbol{u} = \frac{-1}{\sum_{j=1}^{K+1} q_j\left(1-v_j\right)} \left[q_1(1-v_1), \ \dots \ , q_K(1-v_K)\right]^{\mathsf{T}}.$$

and $\boldsymbol{\mathcal{Q}} = \text{diag}(q_i)_{i=1}^{K}$. Its determinant—using basic matrix identities[37]—is thus

$$\begin{aligned}\det(\boldsymbol{\mathcal{J}}) &= \frac{(-1)^K \prod_{j=1}^{K+1} q_j}{\left[\sum_{j=1}^{K+1} q_j\left(1-v_j\right)\right]^K}\left(1 + \mathbf{1}^{\mathsf{T}}\boldsymbol{u}\right) \\ &= \frac{(-1)^K \prod_{j=1}^{K+1} q_j}{\left[\sum_{j=1}^{K+1} q_j\left(1-v_j\right)\right]^{K+1}}.\end{aligned}$$

The joint prior distribution in terms or the vaccine effects is therefore

$$\begin{aligned}&p(v_1, \ \dots \ v_K | n_1(t), \ \dots \ n_K(t), n_{K+1}(t)) \\ &\quad = p(\theta_1(v_1, \ \dots \ v_K), \ \dots \ , \theta_K(v_1, \ \dots \ v_K) | n_1(t), \ \dots \ n_K(t), n_{K+1}(t)) |\det(\boldsymbol{\mathcal{J}})| \\ &\quad = \frac{1}{\text{B}(\boldsymbol{\alpha} + \boldsymbol{n}(t))}\left\{\prod_{l=1}^{K+1}\left[\frac{q_l(1-v_l)}{\sum_{j=1}^{K+1} q_j\left(1-v_j\right)}\right]^{\alpha_l + n_l(t) - 1}\right\}|\det(\boldsymbol{\mathcal{J}})| \\ &\quad = \frac{\prod_{j=1}^{K+1} q_j^{\alpha_j + n_j(t)}\left(1-v_j\right)^{\alpha_j + n_j(t) - 1}}{\text{B}(\boldsymbol{\alpha} + \boldsymbol{n}(t))\left[\sum_{j=1}^{K+1} q_j\left(1-v_j\right)\right]^{n_{\text{total}}(t) + \sum_{j=1}^{K+1}\alpha_j}}.\end{aligned} \tag{A2}$$

Note that since $v_{K+1} = 0$, (A2) defines a K-dimensional, nondegenerate density. Prior to the collection of any data, and when restricting attention to the one-dimensional control comparison, it is straightforward to verify that (A2) reduces to (7).

(b) Without loss of generality, we will prove the result for v_1. to do this, define the following bivariate transformation

$$\begin{cases} v(\theta_1, \theta_{\text{ctrl}}) = 1 - \dfrac{q_{\text{ctrl}}}{q_1}\dfrac{\theta_1}{\theta_{\text{ctrl}}}, \\ u(\theta_1, \theta_{\text{ctrl}}) = \dfrac{q_1}{q_{\text{ctrl}}}\theta_{\text{ctrl}}, \end{cases}$$

whose inverse is

$$\begin{cases} \theta_1(v, u) = u(1-v), \\ \theta_{\text{ctrl}}(v, u) = \dfrac{q_{\text{ctrl}}}{q_1}u. \end{cases}$$

This transformation maps the

$$\{(\theta_1, \theta_{\text{ctrl}}) : 0 \le \theta_1 \le 1, 0 \le \theta_{\text{ctrl}} \le 1 - \theta_1\}$$

support of $p(\theta_1, \theta_{\text{ctrl}} | n_1(t), \ \dots \ , n_{K+1}(t))$ onto

$$\{(v, u) : -\infty < v < 1, 0 \le u \le q_1/(q_1(1-v) + q_{\text{ctrl}})\}.$$

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and has a Jacobian determinant $\det(\boldsymbol{\mathcal{J}}) = \frac{q_{\text{ctrl}}}{q_1}u$. From the basic properties of the Dirichlet distribution,[38] we know that

$$(\theta_1, \theta_{\text{ctrl}}) \sim \text{Dirichlet}\left(\alpha_1, \alpha_{\text{ctrl}}, \sum_{j=2}^{K}\alpha_j\right),$$

and thus the joint posterior density of v and u is given by

$$\begin{aligned}
p(v, u|\boldsymbol{n}(t)) &= \pi(\theta_1(v,u), \theta_{\text{ctrl}}(v,u)|\boldsymbol{n}(t))|\det(\boldsymbol{\mathcal{J}})| \\
&= \frac{[(1-v)u]^{\alpha_1+n_1(t)-1}\left(\frac{q_{\text{ctrl}}}{q_1}u\right)^{\alpha_{\text{ctrl}}+n_{\text{ctrl}}-1}}{\text{B}\left(\alpha_1+n_1(t), \alpha_{\text{ctrl}}+n_{\text{ctrl}}(t), \sum_{j=2}^{K}\left[\alpha_j+n_j(t)\right]\right)}\left[1-\left(1-v+\frac{q_{\text{ctrl}}}{q_1}\right)u\right]^{\sum_{j=2}^{K}\left[\alpha_j+n_j(t)-1\right]}\frac{q_{\text{ctrl}}}{q_1}u \\
&= \frac{\left(\frac{q_{\text{ctrl}}}{q_1}\right)^{\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t)}(1-v)^{\alpha_1}u^{\alpha_1+\alpha_{\text{ctrl}}+n_1(t)+n_{\text{ctrl}}(t)-1}}{\text{B}\left(\alpha_1+n_1(t), \alpha_{\text{ctrl}}+n_{\text{ctrl}}(t), \sum_{j=2}^{K}\left[\alpha_j+n_j(t)\right]\right)}\left[1-\left(\frac{q_1(1-v)+q_{\text{ctrl}}}{q_1}\right)u\right]^{\sum_{j=2}^{K}\left[\alpha_j+n_j(t)\right]-1}.
\end{aligned}$$

The marginal posterior density of r is thus (after a change of variables)

$$\begin{aligned}
p(v|n_1(t), \dots, n_K(t), n_{\text{ctrl}}(t)) &= \int_0^{q_1/(q_1(1-v)+q_{\text{ctrl}})} p(v, u|n_1(t), \dots, n_K(t), n_{\text{ctrl}}(t))\text{d}u \\
&= \frac{q_1^{\alpha_1+n_1(t)}q_{\text{ctrl}}^{\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t)}(1-v)^{\alpha_1+n_1(t)}}{\left[q_1(1-v)+q_{\text{ctrl}}\right]^{\alpha_1+\alpha_{\text{ctrl}}+n_1(t)+n_{\text{ctrl}}(t)}}\frac{\int_0^1 z^{\alpha_1+\alpha_{\text{ctrl}}+n_1(t)+n_{\text{ctrl}}(t)-1}(1-z)^{\sum_{j=2}^{K}\left[\alpha_j+n_j(t)\right]-1}\text{d}z}{\text{B}\left(\alpha_1+n_1(t), \alpha_{\text{ctrl}}+n_{\text{ctrl}}(t), \sum_{j=2}^{K}\left[\alpha_j+n_j(t)\right]\right)} \\
&= \frac{\text{B}\left(\alpha_1+\alpha_{\text{ctrl}}+n_1(t)+n_{\text{ctrl}}(t), \sum_{j=2}^{K}\left[\alpha_j+n_j(t)\right]\right)}{\text{B}\left(\alpha_1+n_1(t), \alpha_{\text{ctrl}}+n_{\text{ctrl}}(t), \sum_{j=2}^{K}\left[\alpha_j+n_j(t)\right]\right)}\frac{q_1^{\alpha_1+n_1(t)}q_{\text{ctrl}}^{\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t)}(1-v)^{\alpha_1+n_1(t)}}{\left[q_1(1-v)+q_{\text{ctrl}}\right]^{\alpha_1+\alpha_{\text{ctrl}}+n_1(t)+n_{\text{ctrl}}(t)}} \qquad \text{(A3)} \\
&= \frac{q_1^{\alpha_1+n_1(t)}q_{\text{ctrl}}^{\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t)}(1-v)^{\alpha_1+n_1(t)}}{\text{B}(\alpha_1+n_1(t), \alpha_{\text{ctrl}}+n_{\text{ctrl}}(t))\left[q_1(1-v)+q_{\text{ctrl}}\right]^{\alpha_1+\alpha_{\text{ctrl}}+n_1(t)+n_{\text{ctrl}}(t)}} \\
&= \frac{q_*^{\alpha_{\text{vax}}+n_{\text{vax}}(t)}(1-q_*)^{\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t)}(1-v)^{\alpha_{\text{vax}}+n_{\text{vax}}(t)-1}}{\text{B}(\alpha_{\text{vax}}+n_{\text{vax}}(t), \alpha_{\text{ctrl}}+n_{\text{ctrl}}(t))(1-q_*v)^{\alpha_{\text{vax}}+\alpha_{\text{ctrl}}+n_{\text{vax}}(t)+n_{\text{ctrl}}(t)}},
\end{aligned}$$

where $q_* = \frac{q_1}{q_1+q_{\text{ctrl}}}$ is the normalized allocation probability to the first vaccine arm. It is easy to verify that (A3) only depends on q_* and $1-q_*$ through their ratio, and we may thus use the original allocation probabilities instead.

(c) Calculating the expected value of a transformed random variable,

$$\begin{aligned}
\mathbb{E}[v_i|\boldsymbol{n}(t)] &= \int_0^1\left\{1-\frac{\theta}{1-\theta}\frac{q_{\text{ctrl}}}{q_i}\right\}\pi(\theta)\text{d}\theta \\
&= 1-\frac{q_{\text{ctrl}}}{q_i}\frac{\int_0^1\theta^{\alpha_i+n_i(t)}(1-\theta)^{\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t)-2}\text{d}\theta}{\text{B}(\alpha_i+n_i(t), \alpha_{\text{ctrl}}+n_{\text{ctrl}}(t))} \\
&= 1-\frac{q_{\text{ctrl}}}{q_i}\frac{\text{B}(\alpha_i+n_i(t)+1, \alpha_{\text{ctrl}}+n_{\text{ctrl}}(t)-1)}{\text{B}(\alpha_i+n_i(t), \alpha_{\text{ctrl}}+n_{\text{ctrl}}(t))} \\
&= 1-\frac{q_{\text{ctrl}}}{q_i}\frac{\alpha_i+n_i(t)}{\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t)-1},
\end{aligned}$$

where the integral converges as long as $\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t) > 1$.

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(d) Since v_i is monotonically decreasing in θ_i, for any $s \in \mathbb{R}$ we have

$$F_{v_i|\boldsymbol{n}(t)}(s) = 1 - F_{\theta_i|\boldsymbol{n}(t)}\left(\frac{q_{\text{vax}}(1-s)}{q_{\text{ctrl}} + q_{\text{vax}}(1-s)}\right),$$

where $F_{v_i|\boldsymbol{n}(t)}(\cdot)$ and $F_{\theta_i|\boldsymbol{n}(t)}(\cdot)$ denote the posterior cumulative distribution functions of v_i and θ_i, respectively, and hence the ξ-th posterior quantile of v_i is given by

$$v_{i,\xi|\boldsymbol{n}(t)} = 1 - \frac{q_{\text{ctrl}}}{q_i}\frac{\theta_{\alpha_i+n_i(t),\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t),1-\xi}}{1-\theta_{\alpha_i+n_i,\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t),1-\xi}}.$$

(e) By the same argument,

$$\begin{aligned} p_i^{\text{effic}}(t) &= \mathbb{P}(v_i > 0 | n_i(t), n_{\text{ctrl}}(t)) \\ &= 1 - F_{v_i|n_i(t),n_{\text{ctrl}}(t)}(0) \\ &= F_{\theta_i|n_i(t),n_{\text{ctrl}}(t)}\left(\frac{q_i}{q_{\text{ctrl}}+q_i}\right) \\ &= F_{\alpha_i+n_i(t),\alpha_{\text{ctrl}}+n_{\text{ctrl}}(t)}\left(\frac{q_i}{q_{\text{ctrl}}+q_i}\right). \end{aligned}$$

APPENDIX B. PROOF OF PROPOSITION 2

(a) Denoting by z_{ijk} the individual (binary) outcome for the kth patient in the jth cluster of treatment arm i, we can calculate

$$\text{Var}\left\{\mathbb{E}\left[z_{ijk}\middle|r_{ij}\right]\right\} = \text{Var}\left[r_{ij}\right] = \mu_i^2/\phi_i,$$

and

$$\begin{aligned} \mathbb{E}\left\{\text{Var}\left[z_{ijk}\middle|r_{ij}\right]\right\} &= \mathbb{E}\left[r_{ij}(1-r_ij)\right] \\ &= \mathbb{E}\left[r_{ij}\right] - \text{Var}\left[r_{ij}\right] - \left\{\mathbb{E}\left[r_{ij}\right]\right\}^2 \\ &= \mu_i(1-\mu_i-\mu_i/\phi_i). \end{aligned}$$

Now, from the law of total variance, we have

$$\begin{aligned} \text{Var}\left[z_{ijk}\right] &= \text{Var}\left\{\mathbb{E}\left[z_{ijk}\middle|r_{ij}\right]\right\} + \mathbb{E}\left\{\text{Var}\left[z_{ijk}\middle|r_{ij}\right]\right\} \\ &= \mu_i(1-\mu_i), \end{aligned}$$

and thus the ICC for the ith treatment arm is given by

$$\rho_i = \frac{\text{Var}\left\{\mathbb{E}\left[z_{ijk}\middle|r_{ij}\right]\right\}}{\text{Var}\left[z_{ijk}\right]} = \frac{\mu_i}{\phi_i(1-\mu_i)}.$$

(b) Taking advantage of (15), we can calculate

$$\begin{aligned} \mathbb{E}\left[\mu_i|\boldsymbol{y}_i,\phi_i\right] &= \frac{\phi_i}{n_i}\mathbb{E}\left[\frac{p_i}{1-p_i}\middle|\boldsymbol{y}_i,\phi_i\right] \\ &= \frac{\phi_i}{n_i}\frac{\int_0^1 \frac{p}{1-p}\cdot p^{a_i+\sum_{j=1}^{K_i}y_{ij}-1}(1-p)^{b_i+K_i\phi_i-1}\text{d}p}{\text{B}\left(a_i+\sum_{j=1}^{K_i}y_{ij}, b_i+K_i\phi_i\right)} \end{aligned}$$

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$$= \frac{\phi_i}{n_i} \frac{\mathrm{B}\left(a_i + \sum_{j=1}^{K_i} y_{ij} + 1, b_i + K_i\phi_i - 1\right)}{\mathrm{B}\left(a_i + \sum_{j=1}^{K_i} y_{ij}, b_i + K_i\phi_i\right)}$$
$$= \frac{\phi_i\left(a_i + \sum_{j=1}^{K_i} y_{ij}\right)}{n_i(b_i + K_i\phi_i - 1)},$$

where the integral converges as long as $b_i + K_i\phi_i > 1$.

APPENDIX C. PROOF OF PROPOSITION 3

For $a_i \in \mathbb{N}^+$,

$$\begin{aligned} \ell_i'(\phi_i) &= \sum_{j:y_{ij}>0} \sum_{m=0}^{y_{ij}-1} \frac{1}{\phi_i + m} - K_i \left\{ \psi\left(a_i + b_i + \sum_{j=1}^{K_i} y_{ij} + K_i\phi_i\right) - \psi(b_i + K_i\phi_i) \right\} \\ &= \sum_{j:y_{ij}>0} \sum_{m=0}^{y_{ij}-1} \frac{1}{\phi_i + m} - \sum_{m=1}^{a_i + \sum_{j=1}^{K_i} y_{ij} - 1} \frac{K_i}{b_i + K_i\phi_i + m}, \end{aligned} \tag{C1}$$

where $\psi(\cdot)$ is the digamma function and the last equality is a result of the relation of its integer differences to the harmonic numbers. If we now rewrite

$$\ell_i'(\phi_i) = f_i(\phi_i) - g_i(\phi_i),$$

for

$$f_i(\phi_i) = \sum_{j:y_{ij}>0} \sum_{m=0}^{y_{ij}-1} \frac{1}{\phi_i + m}$$

and

$$g_i(\phi_i) = \sum_{m=1}^{a_i + \sum_{j=1}^{K_i} y_{ij} - 1} \frac{K_i}{b_i + K_i\phi_i + m},$$

it is easy to verify that $\ell_i'(\phi_i) \xrightarrow[\phi_i \to 0+]{} \infty$ and $\ell_i'(\phi_i) \xrightarrow[\phi_i \to \infty]{} 0$. In addition,

$$\begin{aligned} \lim_{\phi_i \to \infty} \frac{f_i(\phi_i)}{g_i(\phi_i)} &= \frac{\sum_{j:y_{ij}>0} y_{ij}(y_{ij} - 1)}{\left(a_i + \sum_{j=1}^{K_i} y_{ij}\right)\left(a_i + \sum_{j=1}^{K_i} y_{ij} - 1\right)} \\ &< \frac{\sum_{j=1}^{K_i} y_{ij}^2}{\left(\sum_{j=1}^{K_i} y_{ij}\right)^2} \\ &= \frac{\sum_{j=1}^{K_i} y_{ij}^2}{\sum_{j=1}^{K_i} y_{ij}^2 + \sum_{1 \le l < m \le K_i} y_{il} y_{im}} \\ &\le 1. \end{aligned}$$

It follows that there exists a $\phi^* > 0$ such that $\ell_i'(\phi) < 0$ for all $\phi > \phi^*$, and hence $\ell_i'(\phi)$ has at least one root, at which the derivative is trending downwards (and thus a local maximum).

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